



Commodore PET / CBM — Chip Pinouts

MAJOR SOCKETED ICS · CPU · PIA · VIA · CRTC · RAM · TEST & REPAIR

DIP pinouts seen from the top, notch up, pin 1 at top-left. See the **Legend** card for signal notation & the colour key. PET uses the **IEEE-488** bus.

Legend — reading the pinouts notation

/NAME Active LOW — true / asserted when the pin is at logic **LOW** (~0 V). A leading / (datasheets often use an overline) marks these: e.g. /RES /IRQ /CS /WE.

NAME Active HIGH / normal — asserted when the pin is **HIGH** (~+5 V). No slash. R/W is a level: HIGH = read, LOW = write.

Colour key: **Vcc** / **Vdd** = **power** · **GND** / **Vss** = **ground** · **φ** = **clock**. **Orientation:** chip seen from the top, notch up — pin 1 top-left, numbers run **down the left** then **up the right**.

6502 CPU

40-DIP

Vss	1	40	/RES
RDY	2	39	φ2 out
φ1 out	3	38	S0
/IRQ	4	37	φ0 in
NC	5	36	NC
/NMI	6	35	NC
SYNC	7	34	R/W
Vcc	8	33	D0
A0	9	32	D1
A1	10	31	D2
A2	11	30	D3
A3	12	29	D4
A4	13	28	D5
A5	14	27	D6
A6	15	26	D7
A7	16	25	A15
A8	17	24	A14
A9	18	23	A13
A10	19	22	A12
A11	20	21	Vss

PET CPU @ 1 MHz. **Fails:** no startup chirp, blank screen.

6520 PIA x2

40-DIP

Vss	1	40	CA1
PA0	2	39	CA2
PA1	3	38	/IRQA
PA2	4	37	/IRQB
PA3	5	36	RS0
PA4	6	35	RS1
PA5	7	34	/RES
PA6	8	33	D0
PA7	9	32	D1
PB0	10	31	D2
PB1	11	30	D3
PB2	12	29	D4
PB3	13	28	D5
PB4	14	27	D6
PB5	15	26	D7
PB6	16	25	φ2/E
PB7	17	24	CS1
CB1	18	23	/CS2
CB2	19	22	CS0
Vcc	20	21	R/W

PIA1 \$E810: keyboard rows, cassette #1, screen-retrace IRQ, diag sense. **PIA2** \$E820: IEEE-488 data port. **Fails:** dead keyboard (PIA1) / no IEEE bus (PIA2).

6522 VIA

40-DIP

Vss	1	40	CA1
PA0	2	39	CA2
PA1	3	38	RS0
PA2	4	37	RS1
PA3	5	36	RS2
PA4	6	35	RS3
PA5	7	34	/RES
PA6	8	33	D0
PA7	9	32	D1
PB0	10	31	D2
PB1	11	30	D3
PB2	12	29	D4
PB3	13	28	D5
PB4	14	27	D6
PB5	15	26	D7
PB6	16	25	φ2
PB7	17	24	CS1
CB1	18	23	/CS2
CB2	19	22	R/W
Vcc	20	21	/IRQ

\$E840: user port, cassette #2, IEEE-488 handshake, timers, CB2 beep. **Fails:** dead user port / IEEE handshake / no sound.

6545 CRTC video

40-DIP

Vss	1	40	VSYNC
/RES	2	39	HSYNC
LPEN	3	38	RA0
MA0	4	37	RA1
MA1	5	36	RA2
MA2	6	35	RA3
MA3	7	34	RA4
MA4	8	33	D0
MA5	9	32	D1
MA6	10	31	D2
MA7	11	30	D3
MA8	12	29	D4
MA9	13	28	D5
MA10	14	27	D6
MA11	15	26	D7
MA12	16	25	/CS
MA13	17	24	RS
DE	18	23	φ2
CURSOR	19	22	R/W
Vcc	20	21	CCLK

Video timing on the 'universal' & 80-column boards (later PET/CBM). Early 2001s used discrete logic — no CRTC. **Fails:** no / rolling / distorted display.

2114 SRAM 1Kx4

18-DIP

A6	1	18	Vcc
A5	2	17	A7
A4	3	16	A8
A3	4	15	A9
A0	5	14	D1
A1	6	13	D2
A2	7	12	D3
/CS	8	11	D4
GND	9	10	/WE

Static RAM (1Kx4). Early 2001 boards used 6550 instead. **Fails:** garbled screen, memory errors.

ROMs

mask ROM

BASIC	BASIC 1/2/4.0 set
EDITOR	screen editor
KERNAL	I/O + IEEE-488
Character	char generator
24-pin mask ROMs (2316/2332/2364). Fails: won't boot, garbled text.	

Test & Repair

tips

Bus	IEEE-488 (not serial) — monochrome
Black screen	CPU / clock / ROM
Garbage screen	2114 RAM
No display	6545 CRTC (later boards)
Dead keyboard	PIA1
No disk/IEEE	PIA2 / VIA
Sound is a single beep via the VIA's CB2 — there is no sound chip. Socket chips before swapping.	